Figure 8.3 Clocks that move along with galaxies—whose motion, on average, arises only from the expansion of space—provide universal cosmic timepieces. They stay synchronized even though they separate from one another, since they move *with* space but not *through* space.

Notice, of course, that you are free to take your clock, hop aboard a rocket, and zip this way and that across space at enormous speeds, undergoing motion significantly in excess of the cosmic flow from spatial expansion. If you do this, your clock *will* tick at a different rate and you *will* find a different length of elapsed time since the bang. This is a perfectly valid point of view, but it is completely individualistic: the elapsed time measured is tied to the history of your particular whereabouts and states of motion. When astronomers speak of the universe's age, though, they are seeking something universal—they are seeking a measure that has the same meaning everywhere. The uniformity of change throughout space provides a way of doing that.[9]

In fact, the uniformity of the microwave background radiation provides a ready-made test of whether you actually are moving with the cosmic flow of space. You see, although the microwave radiation is homogeneous across space, if you undertake additional motion beyond that from the cosmic flow of spatial expansion, you will not observe the radiation to be homogeneous. Just as the horn on a speeding car has a higher pitch when approaching and a lower pitch when receding, if you are zipping around in a spaceship, the crests and troughs of the microwaves heading toward the front of your ship will hit at a higher frequency than those traveling toward the back of your ship. Higher-frequency microwaves translate into higher temperatures, so you'd find the radiation in the direction you are heading to be a bit warmer than the radiation reaching you from behind. As it turns out, here on "spaceship" earth, astronomers *do* find the microwave background to be a little warmer in one direction in space and a little colder in the opposite direction. The reason is that not only does the earth move around the sun, and the sun move around the galactic center, but the entire Milky Way galaxy has a small velocity, in excess of cosmic expansion, toward the constellation Hydra. Only when astronomers correct for the effect these relatively slight additional motions have on the microwaves we receive does the radiation exhibit the exquisite uniformity of temperature between one part of the sky and another. It is this uniformity, this overall